```
Libraries imported successfully!
```

```
Dataset loaded successfully!
Dataset shape: (4340, 8)
```

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner
0	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Manual	First Owner
1	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Manual	First Owner
2	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Manual	First Owner
3	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Manual	First Owner
4	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Manual	Second Owner

```
Dataset shape: (4340, 8)

Column names: ['name', 'year', 'selling_price', 'km_driven', 'fuel', 'seller_type', 'transmission', 'owner']

Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 4340 entries, 0 to 4339
Data columns (total 8 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   name           4340 non-null   str
 1   year           4340 non-null   int64
 2   selling_price  4340 non-null   int64
 3   km_driven      4340 non-null   int64
 4   fuel           4340 non-null   str
 5   seller_type    4340 non-null   str
 6   transmission   4340 non-null   str
 7   owner          4340 non-null   str
dtypes: int64(3), str(5)
memory usage: 515.2 KB

Statistical summary:
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
name	4340	1491	Maruti Swift Dzire VDI	69	NaN	NaN	NaN	NaN	NaN	NaN	NaN
year	4340.0	NaN	NaN	NaN	2013.090783	4.215344	1992.0	2011.0	2014.0	2016.0	2020.0
selling_price	4340.0	NaN	NaN	NaN	504127.311751	578548.736139	20000.0	208749.75	350000.0	600000.0	8900000.0
km_driven	4340.0	NaN	NaN	NaN	66215.777419	46644.102194	1.0	35000.0	60000.0	90000.0	806599.0
fuel	4340	5	Diesel	2153	NaN	NaN	NaN	NaN	NaN	NaN	NaN
seller_type	4340	3	Individual	3244	NaN	NaN	NaN	NaN	NaN	NaN	NaN
transmission	4340	2	Manual	3892	NaN	NaN	NaN	NaN	NaN	NaN	NaN
owner	4340	5	First Owner	2832	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
Missing values:
```

	Missing Values
name	0
year	0
selling_price	0
km_driven	0
fuel	0
seller_type	0
transmission	0
owner	0

```
Duplicate rows: 763
```

```
Duplicate rows removed: 763
Dataset shape after cleaning: (3577, 8)

Missing values:
```

	Missing Values
name	0
year	0
selling_price	0
km_driven	0
fuel	0
seller_type	0
transmission	0
owner	0

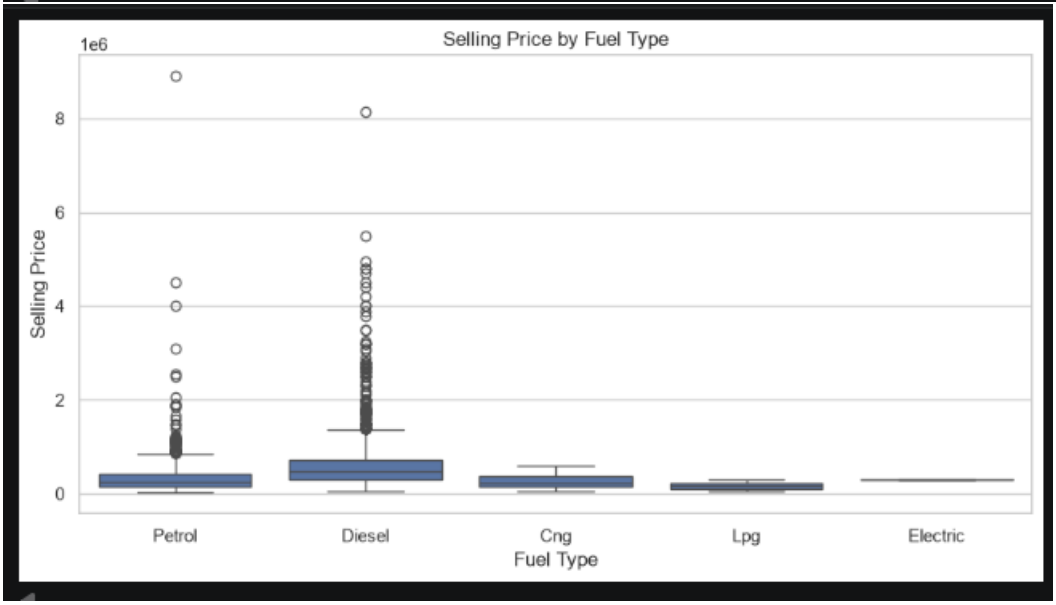
```
Dataset shape after cleaning: (3577, 8)
Remaining missing values: 0
```

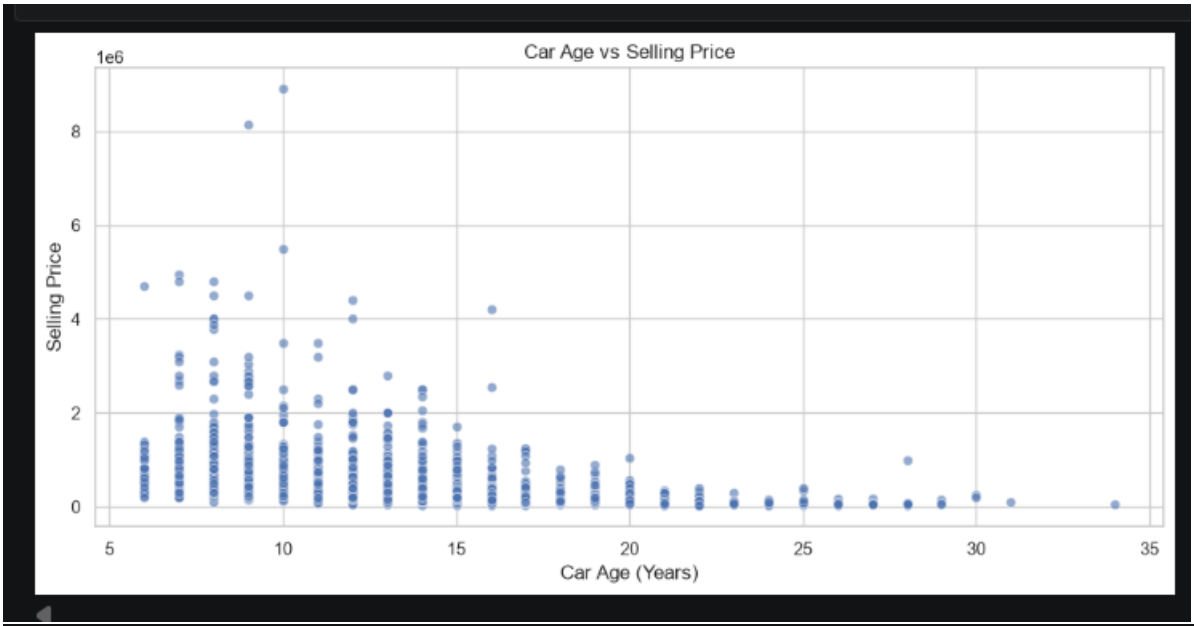
	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner
0	Maruti 800 AC	2007	60000	70000	Petrol	Individual	Manual	First Owner
1	Maruti Wagon R LXI Minor	2007	135000	50000	Petrol	Individual	Manual	First Owner
2	Hyundai Verna 1.6 SX	2012	600000	100000	Diesel	Individual	Manual	First Owner
3	Datsun RediGO T Option	2017	250000	46000	Petrol	Individual	Manual	First Owner
4	Honda Amaze VX i-DTEC	2014	450000	141000	Diesel	Individual	Manual	Second Owner

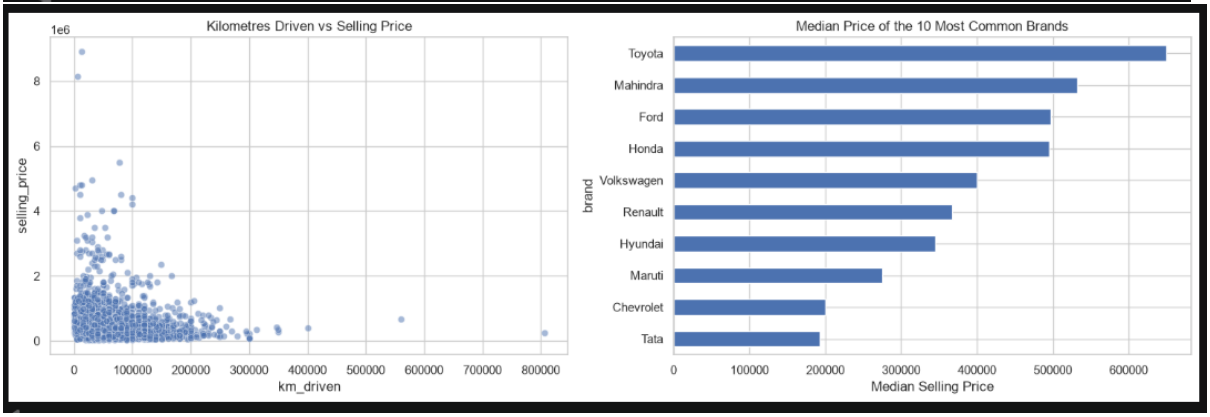
```
Unique brands: 29
```

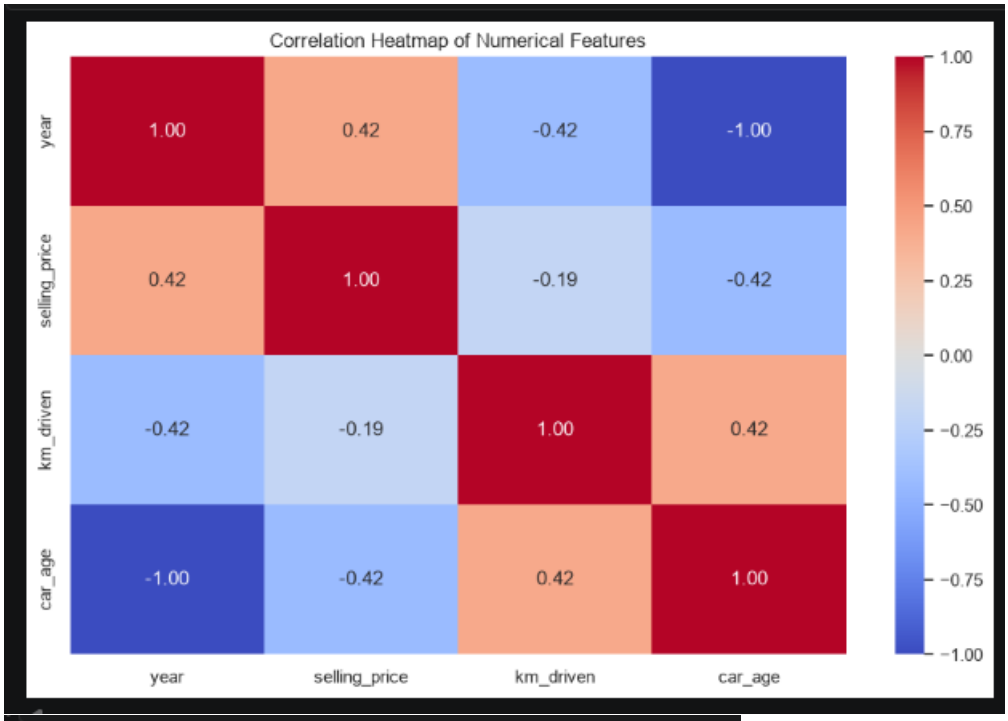
	name	year	brand	car_age	selling_price
0	Maruti 800 AC	2007	Maruti	19	60000
1	Maruti Wagon R LXI Minor	2007	Maruti	19	135000
2	Hyundai Verna 1.6 SX	2012	Hyundai	14	600000
3	Datsun RediGO T Option	2017	Datsun	9	250000
4	Honda Amaze VX i-DTEC	2014	Honda	12	450000






Car Age vs Selling Price
1e6
Selling Price
Car Age (Years)
0
2
4
6
8
5
10
15
20
25
30
35


Kilometres Driven vs Selling Price
1e6
selling_price
km_driven
0
100000
200000
300000
400000
500000
600000
700000
800000
Median Price of the 10 Most Common Brands
brand
Toyota
Mahindra
Ford
Honda
Volkswagen
Renault
Hyundai
Maruti
Chevrolet
Tata
Median Selling Price

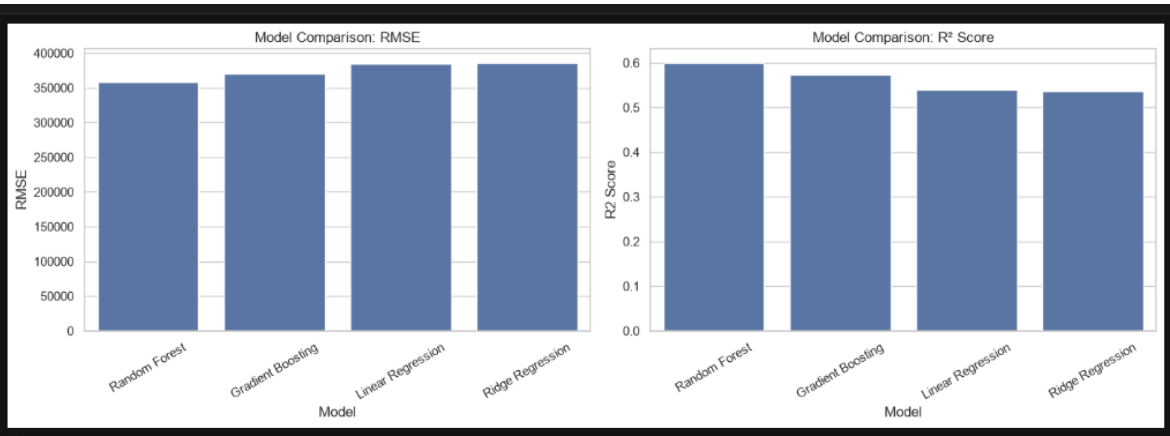



```
Features shape: (3577, 7)
Target shape: (3577,)
```

	car_age	km_driven	brand	fuel	seller_type	transmission	owner
0	19	70000	Maruti	Petrol	Individual	Manual	First Owner
1	19	50000	Maruti	Petrol	Individual	Manual	First Owner
2	14	100000	Hyundai	Diesel	Individual	Manual	First Owner
3	9	46000	Datsun	Petrol	Individual	Manual	First Owner
4	12	141000	Honda	Diesel	Individual	Manual	Second Owner

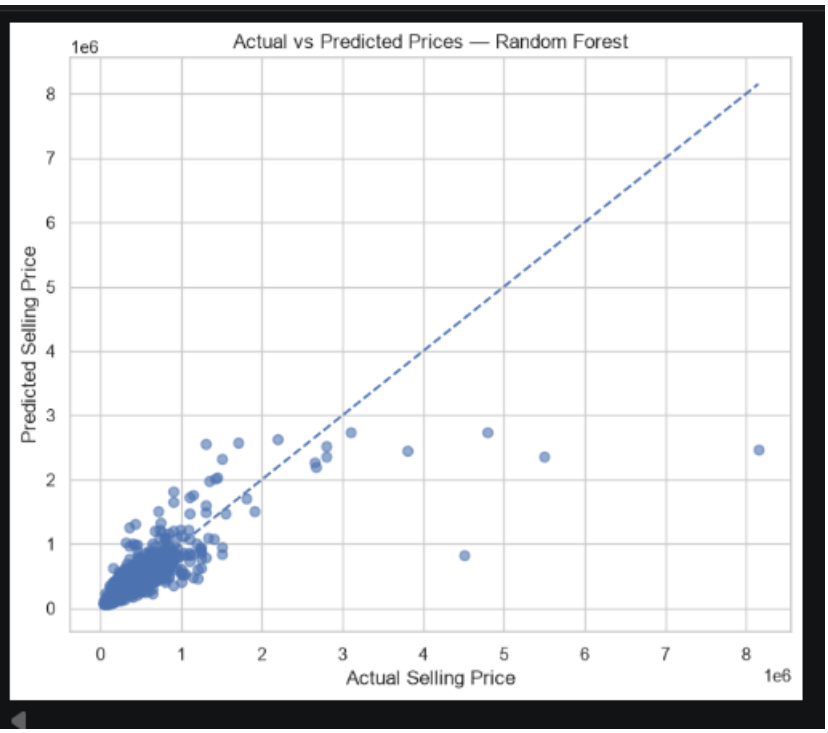
```
Training rows: 2861
Testing rows: 716
```

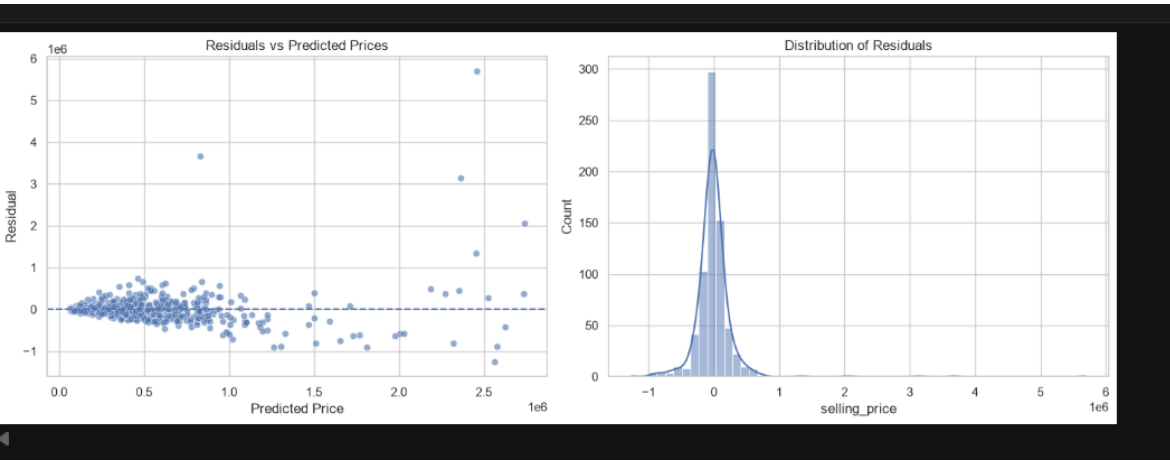
	Model	MAE	RMSE	R2 Score
2	Random Forest	156381.187819	358751.267268	0.600468
3	Gradient Boosting	159394.415842	370774.919211	0.573239
0	Linear Regression	179642.129088	384722.830153	0.540527
1	Ridge Regression	180657.054370	386394.621979	0.536525

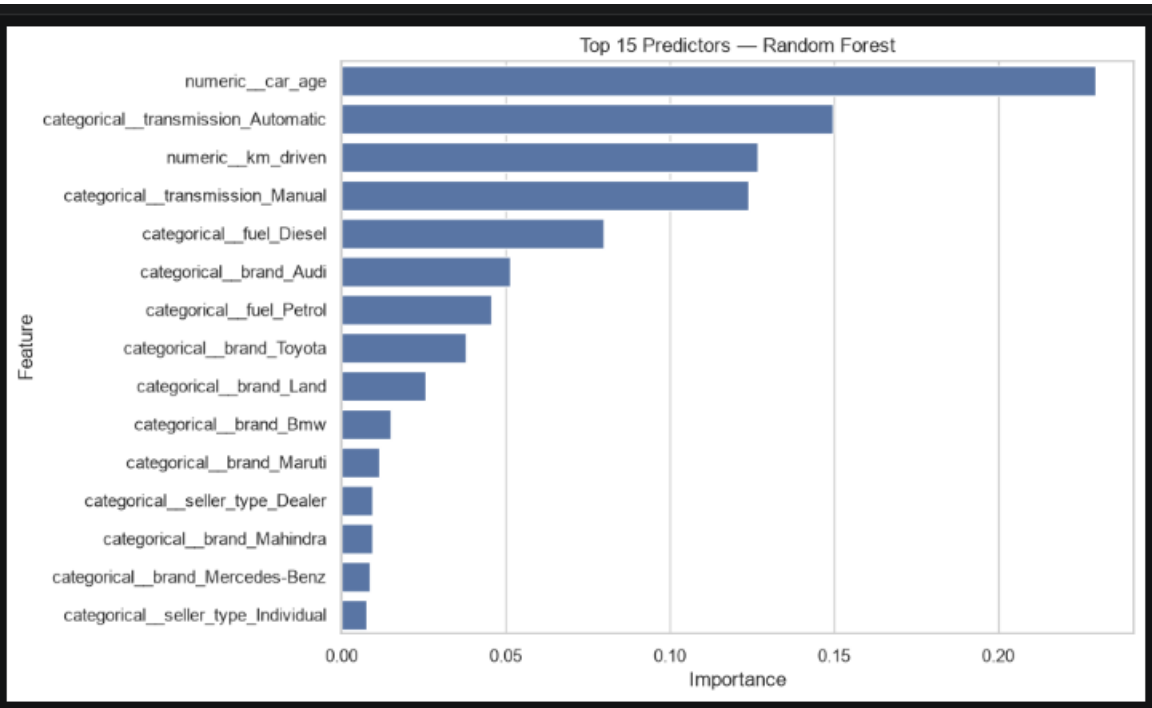


Best model by lowest RMSE: Random Forest

	Model	MAE	RMSE	R2 Score
2	Random Forest	156381.187819	358751.267268	0.600468







	Feature	Importance
0	numeric__car_age	0.229817
36	categorical__transmission_Automatic	0.149668
1	numeric__km_driven	0.126699
37	categorical__transmission_Manual	0.124007
30	categorical__fuel_Diesel	0.079859
3	categorical__brand_Audi	0.051318
32	categorical__fuel_Petrol	0.045607
26	categorical__brand_Toyota	0.037852
15	categorical__brand_Land	0.025620
4	categorical__brand_Bmw	0.014867
17	categorical__brand_Maruti	0.011344
33	categorical__seller_type_Dealer	0.009273
16	categorical__brand_Mahindra	0.009270
18	categorical__brand_Mercedes-Benz	0.008655
34	categorical__seller_type_Individual	0.007553

```
Cross-validation RMSE scores: [271294.96 262490.52 410221.89 234894.36 294942.35]
Mean CV RMSE: 294768.82
RMSE standard deviation: 60839.63
```

```
======================================
FINAL MODEL PERFORMANCE
======================================
Selected model: Random Forest
MAE: 156381.19
RMSE: 358751.27
R² Score: 0.6005
Mean cross-validation RMSE: 294768.82
```

```
Cleaned dataset saved: C:\Users\hp\Desktop\OASIS INFOBYTE\AFTER CODE\3. cleaned_car_price_dataset.csv
Model saved: C:\Users\hp\Desktop\OASIS INFOBYTE\AFTER CODE\best_car_price_model.joblib
```

```
FINAL VERIFICATION
----------------------------------
Dataset shape: (3577, 10)
Missing values: 0
Duplicate rows: 0
Models trained: ['Linear Regression', 'Ridge Regression', 'Random Forest', 'Gradient Boosting']
Best model: Random Forest
Feature importance rows: 15

Basic project checks passed.
```